Supplementary Material: Validation-Based Model Selection and Test Reliability Regret in ADMET Prediction

Huajian Jiang

This supplement records the locked analysis contracts, source-linked result inventory, and evidence boundaries. It is not an independent validation report and does not replace the primary manuscript's stated limitations.

1 Locked experiment matrix

The primary matrix contains 22 endpoints, five model families, and five seeds (1–5), for 550 fit slots. The model families are random forest, XGBoost, LightGBM, GIN, and Chemprop D-MPNN. The cleaned-data sensitivity matrix contains 120 additional fits and is analyzed separately. The descriptive external transport diagnostics contain 120 traditional-model fits, 20 GIN inference files, and 10 Chemprop inference files; these descriptive files are not counted as broad confirmatory external validation. A separate narrow CYP2D6 rerun is reported in the main manuscript and kept in its own audited evidence ledger.

2 Validation-only ExpansionRx replication

The post-hoc ExpansionRx replication contains eight regression endpoints, five candidate model families, and five seeds per family (200 model slots). The validation-reliability selector (VRRS) combines equal within-endpoint rank scores for validation MAE and support/error Spearman correlation. The selector protocol was written after execution and is therefore not preregistered, but the selection table and selector freeze were completed before any test-target directory was opened. The selector map and the validation-MAE-only comparator are:

Endpoint	VRRS choice	Validation-MAE-only choice
`caco2_efflux`	Chemprop D-MPNN	Chemprop D-MPNN
`caco2_papp`	Chemprop D-MPNN	RF
`hlm_clint`	RF	Chemprop D-MPNN
`ksol`	GIN	Chemprop D-MPNN
`logd`	Chemprop D-MPNN	Chemprop D-MPNN
`mbpb`	LightGBM	Chemprop D-MPNN
`mlm_clint`	RF	Chemprop D-MPNN
`mppb`	LightGBM	Chemprop D-MPNN

The VRRS choice differed from the MAE-only choice on 6/8 endpoints. Leave-one- seed stability agreed with the full five-seed VRRS choice in 33/40 comparisons (82.5%). This is a validation-stage

stability diagnostic from a post-hoc rule; no ExpansionRx test targets were read, and no unseen-test improvement or independent external validation is claimed.

3 Primary reliability dimensions

Dimension	Locked definition and interpretation
ECE	Expected calibration error over ten equal-width probability bins for binary tasks.
Brier	Mean squared probabilistic error for binary tasks.
Coverage error	Absolute deviation from nominal 90% residual-interval coverage for regression tasks; validation residuals are used, so this is a diagnostic rather than an independent conformal guarantee.
Relative excess AURC	Risk-coverage area above a loss-ordered descriptive oracle, normalized by mean sample loss.
Applicability alignment	Spearman association between maximum Morgan/Tanimoto support and negative absolute prediction error.

4 Key numeric results

Dimension	Endpoints	Mean regret	95% CI	Practical / tie-robust
ECE	13	0.0325	[0.0148, 0.0539]	9/13; 6/13
Brier	13	0.0092	[0.0034, 0.0169]	6/13; 2/13
Coverage error	9	0.0179	[0.0099, 0.0258]	6/9; 6/9
Relative excess AURC	22	0.0793	[0.0401, 0.1251]	15/22; 12/22
Applicability alignment	22	0.1195	[0.0739, 0.1770]	15/22; 12/22

Table 3: Source-linked summaries from the locked reliability-regret tables.

5 Evidence and limitations

The retained core-matrix release gate reports 550/550 exact run keys and status `GO`. The formal CPU ledger was executed in the tracked remote research environment; the local Mac replay of the 330 traditional-model artifacts is a separate verification lane. The broad confirmatory external-cohort matrix remains unstarted because its outcome-blind power screen is `POWER_FAIL_HOLD`; the planned matrix therefore remains `0/150`, not a completed confirmatory result. A separate narrow CYP2D6 rerun completed 10/10 prediction packages, but it is exploratory only: test-role metrics were examined before protocol freeze, and all ten rerun predictions were byte-identical to the earlier candidate outputs. Hash and arithmetic checks do not repair this prior exposure. The CYP2D6 diagnostic is excluded from confirmatory evidence and is limited to that endpoint and its two source cohorts. Raw TDC rows and model checkpoints are not redistributed in the public source package, so sanitized prediction views alone cannot reproduce label-dependent metrics.

6 Runtime environment and literature boundary

The verified formal neural environment was Linux x86_64 with an NVIDIA A40, CUDA 12.8, Python 3.12.10, Chemprop 2.2.1, PyTorch 2.7.1+cu128, RDKit 2026.09.1, and NumPy 2.5.4. The independent Mac replay used macOS 26.6 arm64, Python 3.11.15, scikit-learn 1.9.1, RDKit 2026.09.1, XGBoost 3.2.0, and LightGBM 4.7.0. The exact package inventory of the formal remote traditional CPU lane was not preserved in the local manifests and is not inferred. The full inventory and lane boundary are recorded in the repository environment inventory document.

The closest cited studies differ in estimand. Zhao et al. evaluate challenge-specific ADMET reliability scenarios; Kamuntavičius et al. study representation/optimization and cross-source transfer; Zheng et al. study a structural-frontier split against a matched scaffold control. The present work freezes one audited TDC snapshot and reports validation-selection versus multi-dimensional held-out reliability regret. The comparison audit is recorded in the repository comparison audit.

7 Source inventory

The manuscript tables, figure manifests, cleaned-sensitivity outputs, cross-fit outputs, and reviewer-defense diagnostics are indexed under the corresponding results directories. The numeric audit and citation audit are copied into this paper directory for release-time hashing. All paths in the public release must remain relative and de-identified.